

Sreehari Sreedev

San Ramon, CA | me@ssree.dev | github.com/sreehax | linkedin.com/in/ssreedev | ssree.dev

EDUCATION

Arizona State University

Bachelor's of Science, Computer Science

Tempe, AZ

Aug 2022 — May 2026

- Enrolled in Barrett, The Honors College
- Relevant Coursework: Discrete Math Structures, Applied Linear Algebra, Data Structures and Algorithms, Database Management, Intro to Theoretical Computer Science

WORK EXPERIENCE

Technical Director

RaRe N Precious

May 2021 — Present

San Ramon, CA

- Increased measured productivity by 75% by automating manual tasks such as Invoice Management
- Wrote in house tax and accounting software using optimal algorithms to achieve the customized needs of the business
- Data is aggregated from several distinct sources to build a unified report needed for State and Federal tax purposes
- Ongoing maintenance work done as needed for adapting to required changes in data schema
- Implemented a system for automatic processing of product images for the company website

Software Engineering Intern

Prognosis

May 2024 — August 2024

Bay Area, CA

- Goal: develop a tool to assist Doctors in reaching a diagnosis by summarizing data
- Analyzed patient similarity, enabling the formation of patient cohorts based on their medical profiles
- Used Node2Vec vector embeddings with cosine search, PyTorch, React.js, Next.js, Flask, and Neo4j

PROJECTS

Founder, Frostium Technologies

April 2024 — Present

- Goal: write an Operating System that ensures maximum security without compromising performance, while being scalable to as many required applications as possible
- Uses a containerized seL4-inspired microkernel architecture
- Linux compatibility in early stages of testing, achieved through userspace service and hypervisor
- Runs on real x86_64 and arm64 hardware
- Geared towards desktop and server use. Embedded usage in development.

Author, Evergreen

June 2024 — Present

- Goal: allow any programmer to use mathematical analysis to prove or disprove that their software behaves correctly
- Simple swift-like programming language with an emphasis on formal verification
- Supports Calculus of Constructions and Inductive Types
- Verification hints are interlaced with code, allowing a programmer to easily build up a specification of their program

EXTRACURRICULAR ACTIVITIES

Security Research

Dec 2019 — Present

- Bypassed Intel BootGuard firmware protections and Intel Management Engine code signing on modern Lenovo ThinkPads
- This vulnerability is in the firmware and completely unpatchable, allowing me to bypass the hardware root of trust
- Found vulnerabilities in the REST backends of many academic-related testing programs, which previously leaked privileged information

Capture The Flag Competitions

Jan 2020 — Present

- Community leader for pwnable.tw, solved many CTF challenges and provided writeups
- Orange Belt on pwn.college, provides sensible help to budding hackers who want to level up their skills

SKILLS AND AWARDS

- **Programming Languages:** AMD64 and ARM64 Assembly, Rust, JavaScript, Python, C/C++, HTML/CSS, Java, Zig
- **Technologies:** Linux, CANBUS, u-boot, Nix, NixOS, gdb/lldb, UNIX, seL4, Docker, NGINX, PyTorch
- **Awards:** NorCal First Tech Challenge Robotics Inspire Award, SolarHacks I 1st Place